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8. Discuss the concept of data independence and explain its importance in a database environment.
9. (a) Discuss the time-stamp ordering protocol for concurrency control. How does strict time stamp ordering differ from basic time stamp ordering?
- (b) What is a certify locks? And what are the advantages and disadvantages of using certify locks.

BCA-501 (N)

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[Total No. of Printed Pages : 4]

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B.C.A. (Fifth Semester)

EXAMINATION, MARCH-2021

(New Course)

Paper First

INTRODUCTION TO DBMS

Time : Three Hours]

[Maximum Marks : 75

Note: Attempt questions from all Sections as directed.

Inst.: The candidates are required to answer only in serial order. If there are many parts of a question, answer them in continuation.

Section-A

(Short Answer Type Questions)

Note : Attempt **all** questions. Each question carries 3 marks.

- 1 (A) What primary characteristics should OID possess?

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- (B) What does the term unnormalised relation refer to?
- (C) What is meant by attribute presentation condition on decomposition?
- (D) Discuss the actions taken by the read-item and write-item operations on a database.
- (E) What is the difference between conflict equivalence and view-equivalence?
- (F) Identify three typical list of transactions that are maintained by recovery subsystem.
- (G) What are the components of the data dictionary?
- (H) Why are relationship between entities important?
- (I) Discuss the problem of dead lock and starvation, and the different approaches to dealing with these problems.

Section-B

Note : Attempt any **Two** question. Each question carries **12** marks.

- 2. What are the software components in client server DBMS ? Compare the two-tier and three-tier architecture.

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- 3. Explain the following with their advantages and disadvantages:
 - (a) Object oriented data model
 - (b) Relational database model
 - (C) Network database model
 - (D) Entity relationship data model
 - (E) Hierarchial database model
- 4. Discuss two multi-version techniques for Concurrency control.
- 5. Explain database storage and database protection.

Section-C

Note : Attempt any **Two** questions. Each question carries **12** marks.

- 6. Prove that any relation schema with two attributes is in BCNF.
- 7. (a) What is dead lock ? Explain deadlock handling techniques with appropriate examples.
(b) How can the database be recovered through shadow paging scheme ?